Table 3. Maldives: Public Sector Debt Sustainability Framework, Baseline Scenario, 2014–37

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2014	2015	2016	Average 5/	Standard Deviation 5/	2017	2018	2019	2020	2021	2022	2017-22 Average	2027	2037	2023-37 Average
Public sector debt 1/	54.2	56.9	65.7			69.4	72.0	73.5	74.3	74.3	73.2		71.8	69.8	
of which: foreign-currency denominated	19.3	17.1	20.8			27.0	28.7	30.4	31.6	32.5	29.7		30.7	27.7	
Change in public sector debt	0.5	2.7	8.8			3.7	2.6	1.5	0.7	0.0	-1.1		-0.2	-0.1	
Identified debt-creating flows	1.5	2.4	7.3			3.9	2.7	1.7	0.8	0.1	-1.0		-0.1	-0.1	
Primary deficit	5.1	4.6	8.2	6.7	4.0	5.7	4.6	3.3	2.5	1.9	0.8	3.1	0.9	0.9	0.9
Revenue and grants	26.7	27.7	28.4			28.7	28.6	28.3	28.4	28.5	28.6		28.8	28.8	
of which: grants	0.3	1.0	0.3			0.3	0.2	0.2	0.2	0.2	0.2		0.1	0.1	
Primary (noninterest) expenditure	31.7	32.3	36.6			34.4	33.2	31.6	30.9	30.4	29.5		29.7	29.7	
Automatic debt dynamics	-3.6	-2.2	-1.0			-1.8	-1.9	-1.6	-1.7	-1.8	-1.8		-1.0	-1.0	
Contribution from interest rate/growth differential	-2.9	-1.5	-0.9			-1.8	-1.9	-1.6	-1.6	-1.6	-1.7		-1.0	-1.0	
of which: contribution from average real interest rate	0.8	0.2	1.2			1.1	1.2	1.7	1.9	1.9	1.9		2.0	2.1	
of which: contribution from real GDP growth	-3.8	-1.7	-2.1			-2.9	-3.1	-3.3	-3.5	-3.5	-3.5		-3.0	-3.1	
Contribution from real exchange rate depreciation	-0.6	-0.7	0.0			0.0	0.0	0.0	-0.1	-0.1	-0.1		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-1.0	0.3	1.5			-0.1	-0.1	-0.1	-0.1	-0.1	-0.1		-0.1	0.0	
Other Sustainability Indicators															
PV of public sector debt	...	...	61.3			64.5	66.4	67.3	67.7	67.3	66.2		64.3	62.9	
of which: foreign-currency denominated	...	...	16.5			22.0	23.1	24.2	25.0	25.5	22.6		23.2	20.8	
of which: external	...	...	16.5			22.0	23.1	24.2	25.0	25.5	22.6		23.2	20.8	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	37.0	40.6	49.4			51.5	48.1	48.0	47.0	46.0	47.1		44.6	45.6	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	216.0			224.8	232.0	237.5	238.4	235.8	231.2		223.2	218.3	
PV of public sector debt-to-revenue ratio (in percent)	...	...	218.2			227.1	234.0	239.4	240.2	237.4	232.6		224.3	218.8	
of which: external 3/	...	...	58.7			77.7	81.3	86.0	88.6	90.0	79.5		80.9	72.4	
Debt service-to-revenue and grants ratio (in percent) 4/	15.9	12.4	12.0			12.8	13.1	14.6	15.3	15.6	25.8		17.7	18.7	
Debt service-to-revenue ratio (in percent) 4/	16.1	12.9	12.1			12.9	13.2	14.7	15.4	15.7	26.0		17.7	18.8	
Primary deficit that stabilizes the debt-to-GDP ratio	4.5	1.9	-0.5			2.0	2.0	1.8	1.8	1.9	1.9		1.1	1.0	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	7.6	3.3	3.9	5.1	4.8	4.6	4.7	4.8	5.0	5.0	5.0	4.8	4.4	4.6	4.5
Average nominal interest rate on forex debt (in percent)	1.6	1.2	1.6	1.9	0.8	2.8	3.1	3.0	2.9	2.9	2.6	2.9	2.5	3.0	2.6
Average real interest rate on domestic debt (in percent)	2.9	0.7	3.1	-0.5	4.2	2.0	2.4	3.4	4.0	3.9	4.3	3.3	4.8	4.5	4.7
Real exchange rate depreciation (in percent, + indicates depreciation)	-2.9	-3.8	-0.1	-3.0	4.6	-0.2	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	4.5	5.3	1.6	7.2	3.7	2.0	2.0	2.0	2.4	2.5	2.5	2.2	2.0	2.0	2.0
Growth of real primary spending (deflated by GDP deflator, in percent)	23.1	5.2	17.8	4.6	8.6	-1.9	1.1	-0.1	2.6	3.3	1.7	1.1	4.4	4.6	4.5
Grant element of new external borrowing (in percent)	...	...	...	...	...	14.4	26.8	27.2	26.8	26.8	28.3	25.1	24.1	15.8	...

Sources: Country authorities; and staff estimates and projections.

1/ Public debt is defined as the debt of the non-financial public sector comprising central government, including loans on-lent to State Owned Enterprises (SOEs), sovereign guarantees issued to SOEs and domestic payable arrears.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4. Maldives: Sensitivity Analysis for Key Indicators of Public Debt, Baseline Scenario, 2017–37

	Projections							
	2017	2018	2019	2020	2021	2022	2027	2037
PV of Debt-to-GDP Ratio								
Baseline	64	66	67	68	67	66	64	63
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	64	68	72	76	79	83	101	131
A2. Primary balance is unchanged from 2017	64	67	71	74	76	79	96	125
A3. Permanently lower GDP growth 1/	64	67	69	71	73	73	85	128
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	64	70	77	79	81	82	89	104
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	64	72	80	79	78	77	74	70
B3. Combination of B1-B2 using one half standard deviation shocks	64	71	78	79	80	79	81	85
B4. One-time 30 percent real depreciation in 2018	64	75	76	75	74	73	70	69
B5. 10 percent of GDP increase in other debt-creating flows in 2018	64	76	76	76	75	74	71	68
PV of Debt-to-Revenue Ratio 2/								
Baseline	225	232	237	238	236	231	223	218
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	225	238	254	267	278	290	350	453
A2. Primary balance is unchanged from 2017	225	236	249	259	268	277	334	433
A3. Permanently lower GDP growth 1/	225	235	245	251	254	256	294	445
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	225	246	271	279	284	286	310	359
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	225	252	281	279	275	268	256	243
B3. Combination of B1-B2 using one half standard deviation shocks	225	249	277	280	279	277	280	295
B4. One-time 30 percent real depreciation in 2018	225	263	267	265	261	254	242	239
B5. 10 percent of GDP increase in other debt-creating flows in 2018	225	264	269	268	264	258	247	236
Debt Service-to-Revenue Ratio 2/								
Baseline	13	13	15	15	16	26	18	19
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	13	13	15	18	21	33	35	51
A2. Primary balance is unchanged from 2017	13	13	15	17	19	31	32	47
A3. Permanently lower GDP growth 1/	13	13	15	16	17	29	26	46
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	13	14	16	19	21	33	28	37
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	13	13	16	23	26	32	21	23
B3. Combination of B1-B2 using one half standard deviation shocks	13	13	16	21	24	33	24	29
B4. One-time 30 percent real depreciation in 2018	13	15	18	19	20	34	24	30
B5. 10 percent of GDP increase in other debt-creating flows in 2018	13	13	16	26	20	31	20	22

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.